Beyond Isolated Events: A Benchmark for Event Relation-Aware Sentiment Quadruple Extraction and Explanation Generation

Anonymous ACL submission

Abstract

Financial events usually contain rich implicit sentiment information, making accurate identification challenging. Existing event-level sentiment analysis predominantly focuses on isolated events, overlooking the impact of inter-event relation on implicit sentiment and lacking investigation into its underlying causes. To this end, we introduce **F**ine-grained **F**inance **E**vent **S**entiment **A**nalysis (FFESA) task from the perspectives of event structure information, event relation, event sentiment and sentiment cause, consisting of three sub-tasks: event sentiment quadruple extraction, event relation extraction, and sentiment explanation generation. To facilitate this research, we construct the fine-grained event-level sentiment dataset FFESA-bench[1], which is annotated via human-in-the-loop interaction. The final dataset is verified by three experts to ensure quality. Additionally, drawing upon Grice's Cooperative Principle, we propose three evaluation metrics for event sentiment explanation, namely sentiment informativeness, factuality, and consistency, to validate the quality of sentiment explanations from multiple dimensions. Finally, we conducted a systematic evaluation of FFESA-Bench based on mainstream large language models. The experimental results reveal the challenges inherent in each sub-task and establish a benchmark for fine-grained financial event sentiment analysis.

1 Introduction

The internet has disseminated large amounts of financial reports texts, which contain abundant event information. The implicit sentiment triggered by these events can impact the entire financial market and provide critical insights for financial decisions (Du et al., 2024). However, relying solely on the implicit sentiment polarity is insufficient to meet the demands of financial decision-making; and revealing the underlying financial logic is also essential

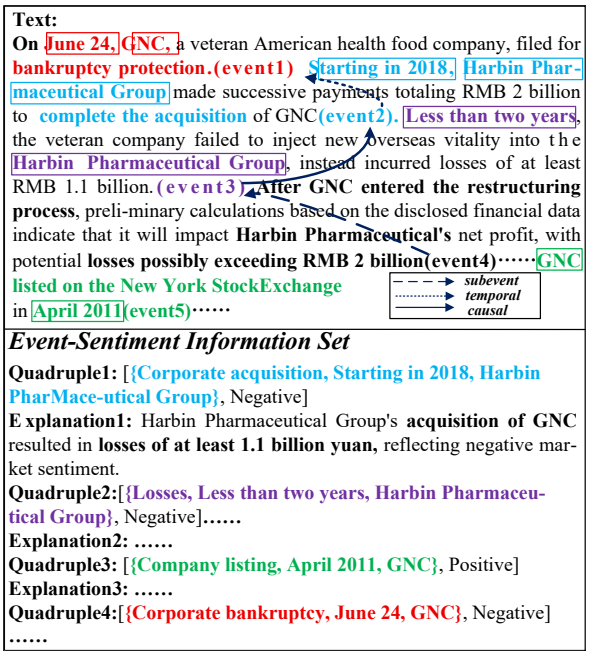


Figure 1: The examples of Fine-grained Event Sentiment Analysis in Financial Text.

(Xing et al., 2020; Luo et al., 2018). Therefore, event sentiment analysis not only focuses on event sentiment polarity, but also explains the underlying causes of event sentiment.

Affective Event Theory (Weiss and Cropanzano, 1996) suggests that sentiment typically stems from specific events. As illustrated in Figure 1, the "company listing" event (event 3) and the "corporate bankruptcy" event (event 4), both triggered by "GNC", exhibit diametrically opposite sentiment polarity. Furthermore, financial event sentiment depends not only on isolated events but also on inter-event relation. For instance, in Figure 1, the "corporate acquisition" event occurring in 2018 (event 1) conventionally carries a positive sentiment; however, due to its causal relationship with the subsequent loss event (event 2) two years later, its sentiment is classified as negative. Moreover, event sentiment explanations can clarify the under-

[1]The dataset will be made publicly available upon acceptance of this paper.

lying causes of the event sentiments. As shown in explanation1 in Figure 1, although event1 is a conventionally positive financial action, it triggered a loss of at least 1.1 billion yuan, rendering its sentiment negative. These cases demonstrate financial texts feature intertwined events and diverse sentiments, posing challenges for event sentiment analysis.

Existing sentiment analysis research mainly includes aspect-level and event-level task: (1) Aspect-level sentiment analysis focuses on aspect-term sentiment (Ding et al., 2024; Lawan et al., 2025) and aspect-category sentiment (Bai et al., 2024; Xu et al., 2025; Wu et al., 2025), excluding event information; (2) Event-level sentiment research focuses on event types (Zhou et al., 2021; Chen et al., 2024; Petrescu et al., 2025) or event structure (Zhang et al., 2022), overlooking inter-event relations and failing to examine the combined impact of event types, information and relations. Furthermore, current sentiment explanation evaluation emphasizes polarity consistency (Zhou et al., 2024; Wang et al., 2025) or factuality verification (Deng et al., 2023), but lacking metrics for event-level assessment and coverage completeness metrics. To address these limitations, we conduct research from four aspects:

Task Definition: We propose **F**ine-grained **F**inance **E**vent **S**entiment **A**nalysis (FFESA) task comprising three interrelated sub-tasks: (1) Event sentiment quadruple extraction: identifying quadruple [event type, subject/object, time, sentiment]; (2) Event relation extraction: identifying causal, temporal, and sub-event relations among events; (3) Event sentiment explanation generation: generating natural language explanations for the event sentiment to uncover the sentiment causes.

Dataset Construction: To support the proposed task, we extend Baidu's DuEE-Fin dataset (Han et al., 2022) to construct FFESA-bench. We select 7,646 texts and devise a "rule guided + human in the loop" workflow to accomplish three annotation tasks: event relations, event sentiment polarities, and event sentiment explanations, thereby ensuring annotation quality and efficiency.

Evaluation Metrics: To address the inadequacy of existing sentiment explanation metrics, we draw inspiration from Grice's Cooperative Principle to design three metrics: Sentiment informativeness, factuality, and consistency, aiming to measure the completeness, reliability, and logical coherence of the event sentiment explanation.

Comprehensive Evaluation: Based on mainstream large language models (LLMs), we design end-to-end and pipeline schemes to evaluate the performance of the three sub-tasks on the FFESA-bench dataset. Experimental results reveal the challenges inherent in each task.

In summary, our main contributions are:

- We propose the task of fine-grained financial event sentiment analysis and construct a dataset comprising 1,328 event relations, 15,189 event sentiment quadruples, and 15,189 event sentiment explanations to provide data resources for implicit event sentiment analysis research.
- We establish multi-dimensional evaluation metrics for sentiment explanations: sentiment informativeness, factuality, and consistency, which enable the assessment of information completeness, reliability, and coherence. Human evaluation confirms their validity.
- We conduct systematic evaluations of mainstream LLMs on FFESA-Bench, revealing bottlenecks in event sentiment quadruple extraction and explanation generation, thereby establishing a benchmark for future research.

2 Related Works

2.1 Sentiment Analysis Dataset

Existing sentiment analysis datasets mainly focus on aspect-level or entity-level sentiment analysis. For example, Tang et al. (2023) proposed a financial entity-level sentiment dataset. Zhu et al. (2025) further introduced a financial entity sentiment dataset with entity location information. Wu et al. (2025) built M-ABSA, a multilingual and multi-domain aspect-based dataset. However, the sentiment granularity in the above studies is limited to aspect terms or entities, making it difficult to capture fine-grained sentiment information.

Building on this, recent datasets have incorporated event information for sentiment analysis. In terms of event types, Chen et al. (2024) proposed extending sentiment analysis to coarse and fine-grained event type hierarchies to predict financial market dynamics. Petrescu et al. (2019) analyzed sentiment tendencies of sudden social media events. Furthermore, other research investigates event structure information on sentiment analysis tasks. Zhou et al. (2021) proposed using event information to analyze the implied sentiment of

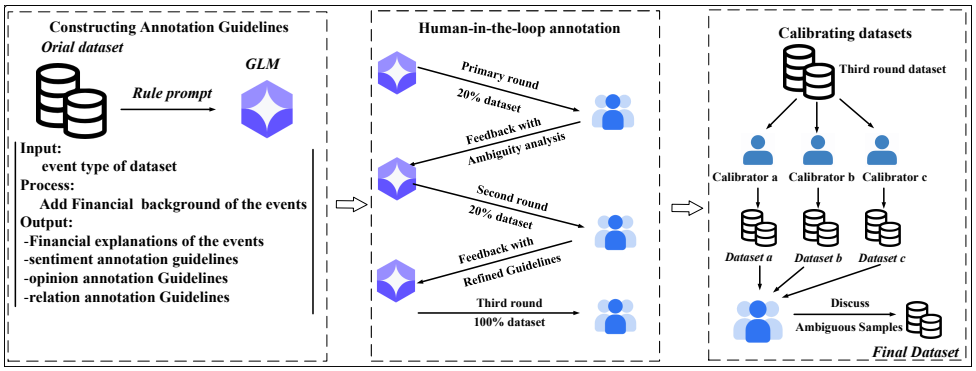


Figure 2: Annotation workflow adopted in FFESA-bench dataset construction.

events. Zhang et al. (2022) leveraged structured event parameter to enhance event-level sentiment analysis. However, these works overlook inter-event relations and lack a comprehensive study on how types, structures, and relations jointly influence sentiment.

2.2 Sentiment Explanation Evaluation

With the advancement of LLMs, the hallucination problem in sentiment explanation generation has become increasingly pronounced. To assess this issue, Deng et al. (2023) proposed assessing sentiment faithfulness; Zhou et al. (2024) evaluated polarity preservation in generated texts; and Wang et al. (2025) measured sentiment consistency. However, current studies predominantly focus on polarity alignment, lacking a systematic evaluation for event-level sentiment granularity.

3 Task Formalization

FFESA task aims to aggregate event sentiment information of document into a unified and interpretable set $L = \{(T_1, P_1) \ldots\ldots (T_i, P_i), R\}$, where T_i denotes event sentiment quadruple, R denotes event relation and P_i denotes explanation.

(1) Event sentiment quadruple extraction

Event tuple extraction: Building upon the financial event extraction task (Han et al., 2022), we formally define the event tuple extraction task. Given a document text $D = \{w_1, w_2, ..., w_n\}$, and get tuple $event_i = [type_i, time_i, subject_i/object_i]$, yielding events set: $Event = \{event_1, event_2, \ldots, event_m\}$.

Event sentiment classification: This task aims to recognize the sentiment polarity of event tuples:

$$sentiment_i = f_{sen}(event_i|D) \quad (1)$$

Where $sentiment_i \in \{pos, neg, neu\}$.

Ultimately, we obtain the event sentiment quadruple: $T_i = [event_i, sentiment_i]$.

(2) Event relation extraction

This task aims to identify the relations $r_{ij} =< event_i, event_j >$ between event tuples in the text, and construct an event relation graph $\mathrm{G} = (\mathrm{E}, \mathrm{R})$, where $R = \{r_{ij}\}$ is the relation set.

(3) Event sentiment explanation generation

This task aims to generate natural language sentiment explanation P_i for event sentiment quadruple, elucidating the underlying causes of the event sentiment, defined as follows:

$$P_i = f_e(T_i|D) \quad (2)$$

4 Dataset Construction

4.1 Dataset Annotation

Building upon Baidu's DuEE-Fin dataset (Han et al., 2022), we selected 7,646 high-quality texts for subsequent annotation. The annotation work comprises three tasks: event sentiment quadruples, event relation, and sentiment explanations. To ensure the standardization and reliability, we designed a "rule guided + human in the loop" workflow. As shown in Figure 2, it constructs three modules:

Constructing Annotation Guidelines: We first utilized Zhipu GLM-4.5 with rule prompts to generate initial annotations for event relation, sentiment polarity and explanations. The annotation team then refined these results through multiple rounds of domain knowledge guided discussion and calibration to produce the preliminary guidelines. Details can be found in the Appendix A.

Human-in-the-loop Annotation: Five finance-background annotators conducted the annotation. To balance quality and efficiency, we adopt a multi-round iterative strategy: First, we employ the LLM to annotate 20% of the data, and then manually

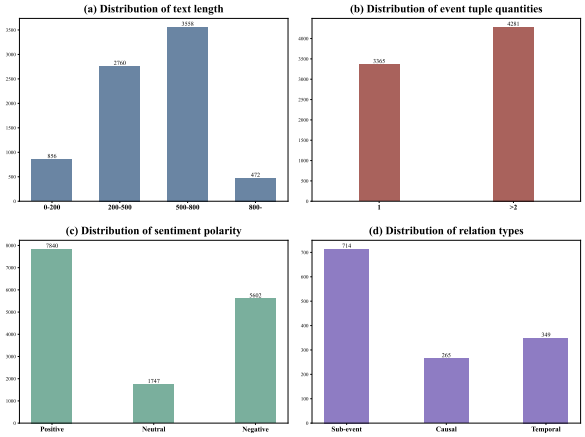


Figure 3: The statistics of distribution of FFESA-bench.

review the results to refine the guidelines by discussing ambiguous cases. Second, we annotate another independent 20% data to further validate and refine the guidelines. Third, with the finalized guidelines, we conduct full-scale annotation on the remaining dataset.

Calibrating datasets: Three independent calibrators (a/b/c) cross-validate the final-round samples, with discrepancies resolved via expert arbitration to yield the gold-standard dataset.Ultimately, Fleiss' Kappa coefficients for event relations, sentiment, and explanations are 0.8395, 0.8536, and 0.8058, demonstrating high annotation reliability.This dataset surpasses existing resources in annotation granularity, as detailed in Appendix B.

4.2 Dataset Distribution

We statistically analyze the dataset from four dimensions: (1)As shown in Figure 3(a), the text length concentrates on the 500–800 character range, accommodating rich event and emotional information; (2) Figure 3(b) shows that the FFESA-bench dataset balances the proportion of single-event tuples and multi-event tuples, providing both benchmark data for single-event and multi-event sentiment research. (3) Figure 3(c) shows balanced positive and negative sentiment distribution; (4) As shown in Figure 3(d), The event relation distribution reveals 1,328 annotated relations covering sub-event, causal, and temporal, providing a logical basis for analyzing implicit event sentiment.

5 Evaluation Framework

For the diverse task formats in section 3, we propose hierarchical evaluation framework for comprehensive and fair assessment.

5.1 Event Sentiment quadruple Evaluation

The event sentiment quadruple extraction task is evaluated using Precision, Recall, and F1. We adopt fuzzy matching via maximum continuous substrings, with a subject, object, or time span considered correct if overlapping the ground truth by at least 70%. Furthermore, to address the inflated recall caused by LLM over-generation, we propose a count capping mechanism: correct matches for any category are capped at the reference count to adjust the True Positive calculation:

$$TP = \sum_{c \in C} \min(N_c^{pred}, N_c^{gold}) \quad (3)$$

Where C denotes the number of categories, $N_c^{gold}$and N_c^{pred} denote the number of predicted tuples and gold tuples for each category.

5.2 Event Sentiment Explanation Evaluation

Inspired by Grice's Cooperative Principle (Grice, 1975), we propose three quantifiable metrics: Sentiment informativeness, Factuality, and Consistency.

5.2.1 Sentiment Informativeness and Factuality

Event sentiment explanations are characterized by expressive diversity and rich emotional reasoning content. Traditional n-gram metrics are limited by lexical matching (Ng and Abrecht, 2015). Therefore, we adopt the event atom as the fundamental evaluation unit (Min et al., 2023) and propose two metrics: sentiment informativeness and factuality.

(1) Event Atoms and Reasoning

In this work, the event atom is the minimal semantic unit preserving consistency across expressions. Thus, each sentiment explanation P_i is decomposed into an event atom set $S_i = \{e_1, e_2, \ldots\ldots e_k\} = Decompose\,(P_i)$. This function can be achieved by LLMs.

To quantify the degree of semantic matching between the predicted and reference sentiment explanations, we define an entailment function:

$$Entail(e, P) = \begin{cases} 1, \text{if } P_i \Rightarrow \text{e} \\ 0, otherwise \end{cases} \quad (4)$$

where e represents an event atom in the predicted sentiment explanation, and P represents the reference sentiment explanation. This function can be implemented using widely-used Natural Language Inference models (Klemen et al., 2024). More details can be found in Appendix C.

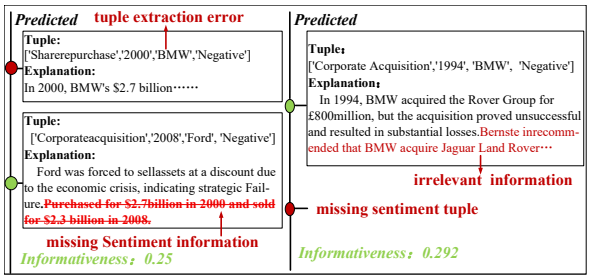


Figure 4: Example of sentiment informativeness.

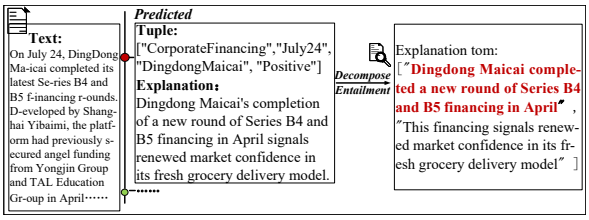


Figure 5: Example of factuality.

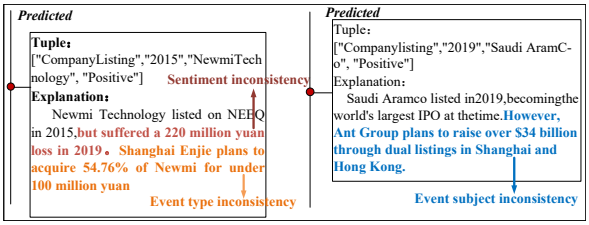


Figure 6: Example of consistency.

Based on this, the predicted reference sentiment explanation P_i and $\hat{P}_i$ are respectively decomposed into atom sets S_i and $\hat{S}_i$. To evaluate the generated explanations, we use the entailment score Ent_p to measure the amount of valuable event information:

$$Ent_p(S_i, \hat{S}_i) = \frac{1}{|\hat{S}_i|} \sum_{\hat{e}_i \in \hat{S}_i} Entail(\hat{e}_i, S_i) \quad (5)$$

Similarly, we can get the entailment recall Ent_r. The entailment Ent_{f1} is calculated as follows:

$$Ent_{f1}(S_i, \hat{S}_i) = \frac{2 * Ent_r * Ent_p}{Ent_r + Ent_p} \quad (6)$$

(2) Sentiment Informativeness

Sentiment informativeness quantifies the completeness of predicted explanations over reference sentiment information, and evaluates alignment at each node. As illustrated in Figure 4, four error categories are identified: (1) tuple extraction error; (2) missing sentiment tuple; (3) missing sentiment information; (4) irrelevant sentiment information.

The above examples show that event tuples affect explanation quality. Thus, evaluation requires the alignment of predicted and reference tuples, following the "align first, quantify later" principle:

First, to evaluate whether the predicted quadruple is correct, we define function as follows:

$$Match(\hat{T}, L) = \begin{cases} 1, \text{if } \hat{T} \in \text{L} \\ 0, otherwise \end{cases} \quad (7)$$

where $\hat{T}$ denotes predicted event sentiment tuple and L denotes reference sentiment line.

To align correctly predicted quadruple with reference explanation, we define function as follows:

$$Align(\hat{T}, L) = \begin{cases} P, if(T, P) \in L and T = \hat{T} \\ null, otherwise \end{cases} \quad (8)$$

where P denotes the reference explanations.

Second, for each predicted explanation $\hat{P}_i$, we define the sentiment informativeness $seninfo$:

$$seninfo((\hat{T}_i, \hat{P}_i), L) = Ent_{f1}(Align(\hat{T}_i, L), \hat{P}_i) * Match(\hat{T}_i, L) \quad (9)$$

Finally, we aggregate the sentiment explanations of the document text to obtain $info$:

$$info(\hat{L}) = \frac{1}{|\hat{L}|} \sum_{(\hat{T}_i, \hat{P}_i)}^{|\hat{L}|} seninfo(L, \hat{L}) \quad (10)$$

(3) Factuality

The factuality metric evaluates explanation truthfulness. Given LLMs' tendency to hallucinate (Ji et al., 2023) and misalign time-events (as shown in Figure 5), factuality detection is crucial (Xu et al., 2024; Setty, 2024). Therefore, we adopt atomic entailment precision to compare generated explanations against the original text, assessing the accuracy of each sentiment information unit. Factuality score is defined as follows:

$$Fact(\hat{L}, A) = \frac{1}{|\hat{L}|} \sum_{\hat{P} \in \hat{L}} Ent_p(\hat{P}, A) \quad (11)$$

where A denotes the original finance text.

5.2.2 Consistency

The consistency metric measures the semantic alignment between explanations and tuples, incorporating three fine-grained consistencies, namely event type (EveC), subject/object (SojC), and sentiment (SenC), as well as overall consistency (AllC). Figure 6 illustrates common consistency issues.

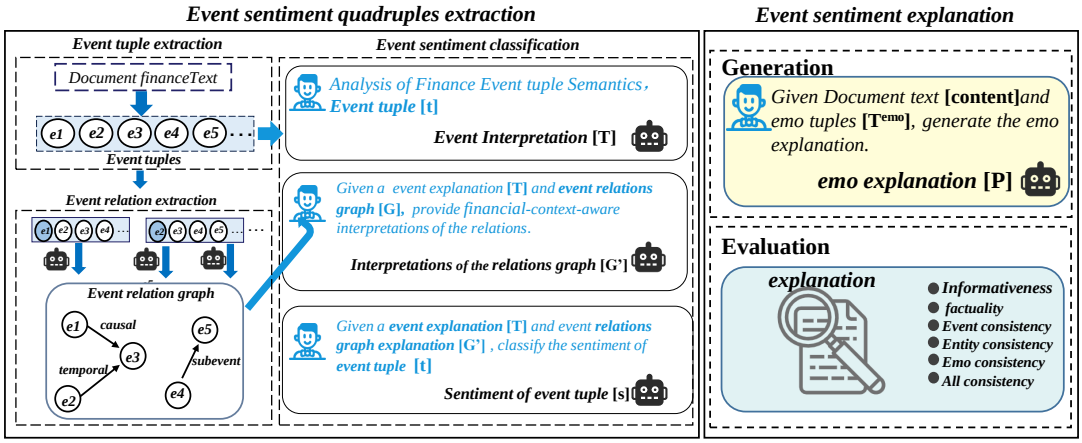


Figure 7: Pipeline for event sentiment tuple extraction and explanation generation

We design an evaluation pipeline inspired by paper review (1) Paraphrasing explanations to reduce bias; (2) Rating three fine-grained consistencies on a 1–3 scale with corresponding rationale; (3) Scoring overall coherence and soundness on a 1–5 scale by synthesizing fine-grained results.

To reduce manual costs, we use Qwen3-Max-Thinking to automate evaluation, leveraging its proficiency in Chinese and annotated examples to ensure reliability. More details are in Appendix D.

6 Experiment

6.1 Experiment Schemes

Given the tasks defined in Section 3, we design two experimental schemes: pipeline and end-to-end.

(1) Pipeline Scheme

The scheme evaluates the performance of each sub-task on the FFESA-bench dataset. As shown in Figure 7, this scheme consists of four modules:

Event tuple extraction: We employ LLMs under zero-shot and 3-shot settings and supervised learning models for event tuple extraction.

Event relation extraction: We employ LLMs under zero-shot and 3-shot settings to evaluate relation extraction performance across three relation types: causal, temporal, and subevent relations.

Event sentiment classification: Motivated by the success of CoT in implicit sentiment analysis (Fei et al., 2023), we adopt a LLM-based Multi-hop Cross-event Sentiment Reasoning (MCSR) as a baseline. The steps of this baseline are as follows:

(a) Expand event tuple **T** via LLMs to generate a semantically enhanced representation **T'**.

(b) Translate relations in **G** into natural language, utilizing LLMs' domain knowledge to enrich background and generate a semantically enhanced **G'**.

(c) Integrate **T'** and **G'** for cross-event reasoning, obtaining polarity **s**, and deriving quadruple T^{emo}.

For event tuples and relations in MCSR, we compare between the predicted and gold label.

Event Sentiment explanation: This module adopts a zero-shot setting to generate explanations, comparing the impact of predicted tuples(PT) and gold tuples(GT) on quality.

(2) End-to-end Scheme

End-to-end joint quadruple extraction is assessed under zero-shot, 3-shot, and fine-tuning settings.

6.2 Experiment Settings

We utilize the following models for evaluation:

Commercial API Models: Three LLms are accessed via API: Qwen-Plus (Yang et al., 2025), Glm-4.5-API (Zeng et al., 2025) and Deepseek-v4-pro-API (Deepseek) (DeepSeek-AI, 2026).

Locally deployed Models: Llama3.1-8B (Dubey et al., 2024), Qwen3-8B (Yang et al., 2025) and Glm4-9B (Zeng et al., 2024).

Supervised Baselines: For event tuple extraction, we adopt ProCNet (Wang et al., 2023) and CAINet (Pan et al., 2024) as baselines.Detailed settings are provided in the Appendix E.

Table 1: Performance of event tuple extraction.

Setting	Model	P	R	F1
Zero-shot	Glm-4.5-API	55.64	37.51	44.81
	Qwen-Plus	50.51	44.88	47.53
	Deepseek	48.36	42.56	45.27
3-shot	Glm-4.5-API	63.27	56.88	59.90
	Qwen-Plus	55.26	48.69	51.77
	Deepseek	55.16	46.89	50.69
Slms	ProCNet	73.17	66.22	69.52
	CAINet	75.68	68.34	71.82

Table 2: Performance of event relation extraction.

Setting	Model	Causal			Subevent			Temporal			Overall
		P	R	F1	P	R	F1	P	R	F1	F1
Zero-shot	Glm-4.5-API	35.8	58.6	44.5	41.9	77.9	54.5	29.7	46.9	36.4	45.0
	Qwen-Plus	28.0	67.4	39.6	34.0	70.5	45.9	23.8	42.6	30.6	41.0
	Deepseek	30.7	43.0	35.9	30.3	62.6	40.9	18.1	45.4	25.9	35.8
3-shot	Glm-4.5-API	49.9	68.9	57.9	51.0	91.4	65.5	30.2	48.5	37.3	53.6
	Qwen-Plus	43.9	60.1	50.8	37.9	90.0	53.4	20.7	41.0	27.5	46.8
	Deepseek	45.2	36.3	40.2	34.6	82.1	48.7	20.5	47.9	28.7	40.8

6.3 Result Analysis

6.3.1 Pipeline Scheme Result

(1) Event tuple extraction

Table 1 presents the event tuple extraction results. (1) Under the zero-shot setting, all models exhibit limited performance, with the best-performing model Glm-4.5 achieving only an F1 of 44.81%; after introducing 3-shot learning, performance generally improves, indicating that contextual examples can effectively enhance LLMs' event tuple extraction capabilities. (2) Compared with supervised models, it is found that although current LLMs demonstrate certain potential under the 3-shot setting, there remains a significant performance gap and considerable room for improvement overall.

(2) Event relation extraction

As shown in Table 2: (1) Overall, Glm-4.5 performs best across both settings, showing in-context examples effectively enhance inter-event understanding. (2) By relation type, sub-event and causal relations show high recall but low precision, suggesting LLMs over-associate similar events due to boundary ambiguity; temporal relations perform worst, implying an over-reliance on surface markers without deep logical reasoning.

(3) Event sentiment classification

Table 3 shows MCSR(Gold) baseline outperforms 3-shot (Gold) by constructing logical associations between events, validating their role in implicit reasoning. However, the F1 of MCSR (Pred) drops sharply, falling below the performance of 3-shot (Gold), highlighting sentiment classification's reliance on information extraction quality.

Furthermore, to examine the impact of different event relations on sentiment reasoning, we compared causal, subevent, and temporal types. The results demonstrate that incorporating event relations consistently improves event sentiment classification across all models. Notably, subevent relations yield the most significant performance gains, whereas causal and temporal relations provide comparable and more limited improvements, suggesting that effectively extracting and leveraging these relation types for reasoning remains a challenge.

Table 3: Performance of sentiment classification.

Setting	Model	P	R	F1
3-shot (gold)	Glm-4.5-API	60.70	59.19	59.94
	Qwen-Plus	52.14	52.10	52.12
	Deepseek	54.87	54.44	54.65
MCSR (gold)	Glm-4.5-API	72.41	66.18	69.16
	Qwen-Plus	60.54	56.53	58.47
	Deepseek	65.45	66.90	66.17
MCSR (pred)	Glm-4.5-API	37.32	31.81	34.35
	Qwen-Plus	33.63	28.88	31.08
	Deepseek	33.62	24.01	28.02

Table 4: performance of F1 on Different Event Relations.

Setting	Model	Causal	Subevent	Temporal
3-shots (Gold)	Glm-4.5-API	21.53	22.19	24.80
	Qwen-Plus	23.40	28.80	20.34
	DeepSeek	18.32	19.07	17.30
MCSR (Gold)	Glm-4.5-API	54.89	61.88	57.53
	Qwen-Plus	38.70	43.12	43.86
	DeepSeek	55.01	66.24	52.19

(4) Event sentiment explanation

Table 5 show that (1) The performance gap between GT and PT indicates that erroneous tuples cause sentiment information loss; (2) Factuality improves most significantly under GT, as high-quality tuples ground reasoning in source text and minimize factual drift. (3)consistency metrics rise, indicating that gold tuple constraints ensure stronger internal unity across event types, subjects/objects and polarities. In summary, high-quality tuples reduce factual errors and inconsistency; and LLMs still exhibit persistent sentiment hallucinations, generating plausible yet ungrounded explanations.

Case Study

Example-1

Bank of China (03988) announced that the US$500 million 0.95% notes due 2023 issued by its Paris branch under the bank's US$4 billion medium-term note program were listed on Euronext Paris on September 22, 2020…

Gold label: [{ "Corporate Financing", "September 22, 2020", "Bank of China" }, "Positive"]

Qwen-plus: [{ "**Company Listing**", "September 22, 2020", "Bank of China" }, "Positive"]

Example-2

High Sun Technology's Subsidiary Proposes to Acquire FinTech Company⋯⋯ High Sun Technology announced that on October 9, 2020, VSP Hong Kong (an indirect non-wholly owned subsidiary of the Company) entered into a memorandum of understanding with the target company regarding the possible investment.⋯⋯

Gold label: [{ "Corporate Acquisition", "VSP Hong Kong", "October 9, 2020" }, "Positive"]

Qwen-plus: [{ " Corporate Acquisition", "**High Sun Technology**", "October 9, 2020" }, "Positive"]

Example-3

……Wabei.net, September 17 news, Kelun Pharmaceutical received a notice on September 16 2019 from Liu Gexin, the Chairman and controlling shareholder of the Company, regarding the release of pledge of some of his shares in the Company…

Gold label: [{ "Release of Pledge", "Liu Gexin", "September 16,2019" }, "Positive"]

Qwen-plus: [{ "Release of Pledge", "Liu Gexin", **September 17**" }, "Positive"]

Example-4

Goertek disclosed on the evening of October 21 that it plans to repurchase shares for CNY 500 million to 1 billion, with the funds intended for an employee stock ownership plan (ESOP). The market suspects this buyback is actually a "pump" to facilitate the major shareholder's sell-off.

Gold label: [{ "Share Repurchase", " Goertek ","Evening of October 21" }, "Negative"]

Qwen-plus: [{ "Share Repurchase", "Goertek" , "Evening of October 21" }, "**Neutral**"]

Figure 8: Examples include text, gold quadruple, and prediction, with red font denoting prediction errors.

Table 5: Performance of sentiment explanation and comparison between Gold Tuple (GT) and Predictions (PT).

Setting	Model	Info	Fact	EveC	SojC	SenC	AllC
PT	Glm4-9b	20.34	60.34	2.13	2.26	2.26	2.71
	Glm-4.5-API	25.28	71.50	2.92	2.55	2.75	3.71
	Qwen3-8b	21.25	61.49	2.74	2.37	2.43	3.57
	Qwen-Plus	22.78	76.12	2.80	2.51	2.70	3.64
	Llama3.1-8b	20.98	60.28	2.43	2.35	2.38	3.02
GT	Glm4-9b	43.96	90.47	2.79	2.32	2.37	2.91
	Glm-4.5-API	52.53	89.00	2.94	2.84	2.84	3.85
	Qwen3-8b	48.16	90.39	2.80	2.89	2.70	3.70
	Qwen-Plus	52.23	90.03	2.80	2.90	2.81	3.80
	Llama3.1-8b	48.94	90.61	2.78	2.57	2.40	3.21

Table 6: This table presents the performance of end-to-end joint event sentiment quadruple extraction.

Setting	Model	P	R	F1
Zero-shot	Glm-4.5-API	18.19	21.29	19.62
	Qwen-Plus	25.86	25.53	25.69
	Deepseek	32.49	31.98	32.24
3-shot	Glm-4.5-API	42.22	40.69	41.44
	Qwen-Plus	35.11	41.65	38.10
	Deepseek	36.36	38.55	37.42
Fine-tuning	Glm4-9b	49.82	47.77	48.78
	Qwen3-8b	75.29	64.62	69.55
	Llama3.1-8b	63.02	56.30	59.47

6.3.2 End-to-end Scheme Result

Table 6 presents end-to-end results: (1) Under the zero-shot setting, joint extraction F1 is only 32.24%, indicating LLMs struggles with multi-event and multi-sentiment texts without example guidance; 3-shot learning raises the score to 41.44%, but challenges remain. (2) Full fine-tuning achieves 69.55%, significantly outperforming the zero-shot and 3-shot settings and demonstrating the value of our high-quality annotations for the task.

6.4 Metrics Alignment with Human

To validate consistency between human and automatic evaluation, we randomly sampled 500 instances annotated by NLP experts. Pearson correlation analysis shows that informativeness and factuality scores reach 78.65% and 98.65%, with all consistency metrics exceeding 95.95%. The results demonstrate the evaluation's reliability and validity. Details can be found in the Appendix F.

7 Case Study

To identify LLM limitations in event sentiment tuple extraction, we identify four distinct error type (Figure 8): (1) Type confusion: high semantic similarity leads to categorization error(Example-1), (2) Subject misidentification: the model prefers frequency entities in multi-entity contexts (Example-2), (3) Temporal misalignment: publication time is frequently mistakenly extracted as event time (Example-3), (4) Sentiment misjudgment: literal keywords override implicit context (Example-4).

8 Conclusion

We introduce the fine-grained event-based financial sentiment analysis task, construct the FFESA-bench dataset and three metrics to detect sentiment explanation hallucinations, namely sentiment informativeness, factuality, and consistency. We investigate LLM performance bottlenecks across subtasks through end-to-end and pipeline benchmarks. In future work, we plan to diversify the language sources and propose more effective LLM-based methods for financial event sentiment analysis.

Limitations

Financial data annotation faces inherent challenges, such as dense domain-specific terminology and high context dependence, rendering dataset construction labor-intensive. Future work should focus on developing efficient annotation pipelines to facilitate the expansion of fine-grained event sentiment datasets. To evaluate the FFESA task, we rely on large language models' APIs, which are costly and may not be accessible to all researchers. Due to the high API costs, we report the results from a single run. Furthermore, since the dataset used in this study is exclusively in Chinese, future research should explore more effective data collection and evaluation methods to extend the applicability to multilingual and multi-domain scenarios.

Ethics Statement

All data in the FFESA-bench dataset are sourced from the DuEE-Fin dataset (Han et al., 2022), which is an open-source dataset available for academic research. Our annotation process was carried out by postgraduate students from Chinese universities serving as annotators, and they received appropriate compensation for their work.

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A Details of the Annotation Workflow

A.1 Annotation Prompts

To bootstrap the annotation process efficiently, we utilized Zhipu GLM-4.5 to conduct preliminary analysis for each event type , aiming to grasp the macroscopic distribution of event sentiment polarities(figure 9) and event relations(figure 10). To mitigate hallucinations, we required the LLM to provide supporting rationales and explanations during relation extraction and sentiment discrimination, thereby facilitating subsequent manual verification. Leveraging these interpretable outputs, experts from both the financial domain and NLP sentiment analysis convened to discuss the results, ultimately constructing the initial annotation guidelines.

A.2 Annotation Guidelines

Taking "Corporate Acquisition" as an example, the annotation rules vary by role. For the acquirer, successful acquisitions or progress updates are labeled as positive, while obstacles, regulatory inquiries, or post-acquisition losses are labeled as negative. For the acquiree, premium acquisitions, stock price increases, or mutual efforts to advance the deal are labeled as positive, whereas hostile takeovers or other adverse scenarios are labeled as negative. In the absence of such explicit information, the label is neutral.

A.3 Ambiguous Examples

Event Type: Corporate Acquisition
Target Event: ["Corporate Acquisition", "2008", "Ford Motor"]
Correct Polarity: Negative

In the original text, "*...in 2000, BMW sold Rover, which owned the Land Rover brand, to Ford for $2.7 billion...in 2008, Ford, hit by the economic crisis, sold it to India's Tata Motors for $2.3 billion*," the sentiment polarity of this event is clearly negative. However, during the initial annotation phase, LLMs and some annotators tended to mislabel it as positive due to insufficient contextual association.

In fact, through cross-sentence numerical comparison, it is evident that Ford's selling price in 2008 ($2.3 billion) was significantly lower than its acquisition price in 2000 ($2.7 billion). The asset depreciation is clear; therefore, contextual reasoning is essential to arrive at the correct negative judgment.

B Dataset Resources

B.1 Comparison of Resources

Compared with existing datasets, FFESA-Bench provides finer-grained event sentiment annotations. As shown in Table 7, it is currently the only resource that simultaneously encompasses event annotation, event relations, sentiment explanations, and event sentiments. pifont

B.2 Dataset Release

Upon acceptance of the paper, FFESA-Bench will be publicly released in the standard JSONL format, containing sentiment quadruples, event relations, and corresponding sentiment explanation fields.

C Event Atoms Decomposition and Reasoning

Event atom-based evaluation metrics require separate decomposition of reference event emotion explanations and generated event emotion explanations. Specifically, event atoms in reference event emotion explanations are pre-annotated by human annotators, while decomposition of generated event emotion explanations is performed using Qwen-Plus for real-time automatic decomposition.

Financial Event Sentiment Analysis Prompt

You are financial event sentiment analysis expert
##Task Overview
Analyze sentiment polarity of financial events in text documents with precision and domain expertise.
##Core Tasks
1. **Sentiment Polarity Analysis**: Classify each event tuple as positive/negative/neutral
2. **Evidence-based Explanation**: Provide sentiment explanation strictly based on original text
3. **Confidence Scoring**: Assign confidence score (0-1 scale)
Analysis Principles
The {Analysis principles} were established by financial domain experts and NLP sentiment analysis experts and define sentiment analysis guidelines for each event type. Corporate acquisition is provided as an example:
For the Acquirer Company
Positive Sentiment Markers: mark as positive if the acquisition completes successfully with mutual benefits, strategic alignment, favorable pricing, synergies, or positive market reaction;
Negative Sentiment Markers:mark as negative if it fails, faces regulatory rejection, overpricing concerns, financial strain, integration risks, or negative market response.
Other: Cases not fitting above categories
For the Target Company
Positive Sentiment Markers： Mark as positive if shareholders receive premium prices, better development opportunities, job security, or strategic benefits;
Negative Sentiment Markers: Mark as negative if the price is undervalued, core business is threatened, layoffs expected, or forced sale due to distress.
Other: Cases not fitting above categories
Consider stakeholder perspectives, market signals, management statements, and acquisition stage when making judgments.
Output Format
```
{
  "sentiment": "positive/negative/neutral",
  "explanation": "Sentiment explanation based on the original text",
  "confidence_score": "0-1"
}
```
Input
Text: {Finance document text}
Event tuple:<triple>

Figure 9: Initial prompts for event sentiment annotation and explanation generation in the preliminary phase.We show examples for the model to better comprehend the task.

You are an expert proficient in event logic modeling.

System Prompt
Task
Your task is to analyze the evolutionary logic and hierarchical relationships between multiple events in a given financial text. Please annotate the provided event pairs (such as E1 and E2) from the following three dimensions: subevent relationship, causal relationship, and temporal relationship.
Output
Return a standard JSON object with the following structure. If no relationship exists, the relations array should be empty:
```
{
  "relations": [
    {
      "head_id": "EX",
      "tail_id": "EY",
      "relation_type": "subevent/causal/temporal",
      "evidence": "Brief rationale"
    }
  ]
}
```
##Examples
(1)Subevent Relationship Example:
E1[Annual loss of 1 billion] and E2[Q2 loss of 300 million]
Relation: subevent
Evidence: E2 represents a specific quarterly component that constitutes the annual loss described in E1
(2)Causal Relationship Example:
E1[Excessive acquisition costs] leads to E2[Unsatisfactory post-acquisition benefits resulting in losses]
Relation: causal
Evidence: High acquisition costs directly caused the subsequent financial losses
(3)Temporal Relationship Example:
E1[Original chairman's term expiration] followed by E2[Election of new chairman]
Relation: temporal
Evidence: Events follow chronological business process sequence without dependency or causation
Input
Financial text {Text}
Containing event set :{Events}

Figure 10: Initial prompts for event relation extraction in the preliminary phase. We show examples for the model to better comprehend the task.

Table 7: Comparison of FFESA-Bench with existing financial event-sentiment datasets.

Dataset	Event Annotation	Event Relation	Sentiment Explanation	Event Sentiment
FinEntity	✗	✗	✗	✗
DuEE-Fin	✓	✗	✗	✗
EFSA	✓	✗	✗	✓
FFESA-Bench	✓	✓	✓	✓

C.1 Manual Decomposition Process

Event Atom Decomposition Process:

(1) Context Understanding: Carefully read the entire event sentiment explanation to understand the main event or narrative described, then identify the primary subject/object and action/state in the sentence.

(2) Event Atom Decomposition: Break down complex sentences into event atom units based on independent actions/states.This approach is demonstrated by breaking down the explanation "Dingdong Mall completed a new round of B4 and B5 financing in April, which indicates that the market has regained confidence in its new fresh food delivery model" into two coherent event units:

- Tom 1: "Dingdong Maicai completed a new round of Series B4 and B5 financing in April"
- Tome 2: "This financing signals renewed market confidence in its fresh grocery delivery model"

C.2 Atomic Decomposition

We employed Qwen-Plus for fully automatic atomic decomposition, with the specific implementation details presented in figure 11.

C.3 Reasoning

We employ Qwen3-Max-Thinking, which demonstrates superior performance in Chinese language processing, as the core tool for entailment judgment.

D Details of the consistency

The evaluation process consists of two steps. In the first step, evaluators score three fine-grained consistencies, as shown in figure 12, figure 13, and figure 14. They must rate each aspect on a 1-to-3 scale and provide required rationales to prevent arbitrary scoring. The second step involves an overall evaluation, where evaluators synthesize the scores and rationales from the first step to assign an overall score on a 1-to-5 scale. This overall score assesses the coherence and logical soundness of the entire explanation, with the evaluation prompt shown in Figure 15. Together, these steps form a complete pipeline from detailed inspection to overall assessment.

Case Study

Quadruple: [“Company Listing”,“2019”, “Saudi Aramco”, “Positive”]

Explanation: Saudi Aramco successfully went public in 2019, marking the world’s largest IPO at the time and a significant milestone in Saudi Arabia’s economic transformation. However, Tesla also built a Gigafactory in Shanghai that same year; its stock price soared, with market capitalization breaking 100 billion USD, making it the biggest winner in the capital markets.

LLM Error Assessment: The opinion information explicitly mentions Saudi Aramco’s 2019 listing with a positive sentiment, which is fully consistent with the quadruple and introduces no irrelevant information.

Error Attribution Analysis: The text is actually only partially consistent. Due to local matching bias, the LLM focused solely on the target entity information in the first half while ignoring the irrelevant financial entity (Tesla) and its sentiment descriptions in the second half, resulting in an inflated score.

Solution: By refining and strictly enforcing the evaluation prompts, the model’s ability to filter out irrelevant entity interference was significantly enhanced, thereby largely resolving this type of evaluation bias.

E Experimental Setup Details

To ensure experimental reproducibility and transparency, we provide detailed descriptions of the experiment.

Atomic Decomposition Prompt

System Prompt

Task
As a Financial Sentiment Analysis Expert, you will receive an event sentiment explanation in the financial domain. This event sentiment explanation may contain descriptions of multiple events as well as different viewpoints. Your task is to decompose this opinion information into simpler atomic propositions or event descriptions. Each atomic proposition or event description should be clear, concise, and accurately reflect a specific event or viewpoint in the original text. Please write each decomposed result as an independent sentence.

Output
Strictly follow the format below, no extra characters allowed.
Return a list of sentences in the format: ['Sentence 1', 'Sentence 2', ...]
}

Example
Ford was forced to sell assets at a discount due to the economic crisis, purchased for $2.7 billion in 2000, sold for $2.3 billion in 2008, the text implies strategic failure.
Output: ['Ford was forced to sell assets at a discount due to the economic crisis', 'Ford purchased the assets for $2.7 billion in 2000', 'Ford sold the assets for $2.3 billion in 2008', 'The text implies Ford's strategic failure']

Input

Event sentiment explanation **{explanation}**

Figure 11: Prompts for atomic decomposition. We show examples for the model to better comprehend the task.

E.1 API Model Versions

The specific versions and access dates of the APIs we used are as follows:

- **DeepSeek-v4-pro**: Accessed on 2026-04-24
- **Qwen-Plus**: Accessed on 2025-12-01
- **GLM-4.5-API**: Accessed on 2025-07-28

E.2 Fine-tuning Hyperparameters

For locally deployed models, all fine-tuning procedures were executed via the LLaMA-Factory framework. Rather than performing task-specific hyperparameter searches, we adopted its standardized default configuration (covering learning rate, batch size, warmup steps, etc.) uniformly across all runs.

E.3 Dataset Splitting Strategy

Prior to any model training or evaluation, the complete corpus was randomly partitioned into training, validation, and test sets at a ratio of 8:1:1. To facilitate experimental reproducibility, the resulting splits will be released in our repository in the standard train/dev/test format, ensuring that the data preprocessing pipeline remains fully transparent and reusable.

F Details of Human Evaluation

The evaluation work invited three experts with extensive experience in the fields of finance and natural language processing, and expertise in sentiment analysis, to participate. The experimental dataset consists of 500 samples, encompassing financial texts, event sentiment quadruples, and predicted explanations. To ensure objectivity and representativeness, samples were selected via stratified sampling, maintaining the original sentiment category distribution.

The evaluation process adhered to pre-established criteria. First, the three evaluators independently conducted the initial scoring for all samples; subsequently, the evaluation team held a discussion session to conduct an analysis and negotiation regarding samples with significant scoring discrepancies or ambiguities; finally, based on the discussion results, the evaluators performed a second round of annotation to obtain the corrected final scores.

As shown in Figure 16, the evaluation centered on seven core dimensions: sentiment informativeness P, sentiment informativeness R, factuality, event type consistency, event Subject/Object consistency, event sentiment consistency, and overall consistency. Specific scoring criteria were established for each dimension to ensure the reliability and comparability of the evaluation results.

Event type Consistency Evaluation Prompt

System Prompt

Task
You are a rigorous reviewer responsible for evaluating the "event type consistency" between 【tuple event】 and 【sentiment explanation】.
Please strictly score the input according to the following rules and provide an evidence.
Core Principles:
(1)Main focus: Whether the sentiment description in the sentiment explanation closely revolves around the tuple event.
(2)If the sentiment information extends to company or market reactions closely related to the tuple event, it is considered relevant supporting information and will not reduce the score.
(3)Only when events completely unrelated to the sentiment of the tuple event appear, should the score be reduced.
Scoring rules:
3 = Excellent The sentiment description in the sentiment explanation is completely consistent with the tuple event and expressed comprehensively; no events unrelated to the sentiment of the tuple event are introduced.
2 = Good The sentiment explanation includes the tuple event with complete sentiment description; however, it simultaneously introduces other events unrelated to the sentiment of the tuple event.
1 = Poor The sentiment explanation mentions the tuple event, but the sentiment description of this event is incomplete or vague; and introduces other events unrelated to the sentiment of the tuple event.
0 = Irrelevant/Error The sentiment explanation does not mention the tuple event; or the event in the sentiment explanation is completely inconsistent with the tuple event.
Output Format:
The output must use JSON format, with a fixed format of:
{"score": X, "evidence": "Your explanation"}
Example:
Example 1:
Quadruple: ["company listing", "2015", "Newmi Technology", "positive"] Tuple Event: company listing
Sentiment Explanation: Newmi Technology was listed on the NEEQ in 2015.
Output: {{"score": 3, "evidence": "The sentiment explanation accurately covers the tuple event (company listing), with consistent sentiment and complete expression, without any sentiment content unrelated to this event."}}
Example 2:
Quadruple: ["company listing", "2015", "Newmi Technology", "positive"] Tuple Event: company listing
Sentiment Explanation: Newmi Technology was listed on the NEEQ in 2015. ENJET plans to acquire 54.76% of Newmi Technology's shares, with the expected transaction cost not exceeding 100 million yuan.
Output: {{"score": 2, "evidence": "The sentiment explanation fully mentions the tuple event and accurately conveys its positive sentiment, but simultaneously introduces the acquisition event, which is unrelated to the company listing, resulting in a score reduction."}}
Example 3:
Quadruple: ["company listing", "2015", "Newmi Technology", "positive"] Tuple Event: company listing
Sentiment Explanation: ENJET plans to acquire 54.76% of Newmi Technology's shares, with the expected transaction cost not exceeding 100 million yuan.
Output: {{"score": 1, "evidence": "The sentiment explanation does not cover the core content of the tuple event (company listing), but only describes the acquisition event, and introduces sentiment events unrelated to the tuple, resulting in a low score."}}
Example 4:
Quadruple: ["company listing", "2015", "Newmi Technology", "positive"] Tuple Event: company listing
Sentiment Explanation: Tesla plans to expand its Gigafactory in Shanghai.
Output: {{"score": 0, "evidence": "The sentiment explanation does not mention the tuple event, and the company listing is completely inconsistent with the content."}}

Input

Event Sentiment Quadruple :{tuple} Tuple event type: {tuple[0]}

Event Sentiment Explanation: {explanation}

Figure 12: Prompts for event type consistency evaluation. We show examples for the model to better comprehend the evaluation.

Event Sentiment Consistency Evaluation Prompt

System Prompt

Task
You are a rigorous reviewer responsible for evaluating the "sentiment consistency" between 【quadruple】 and 【sentiment explanation】.Please strictly score the input according to the following rules and provide an explanation.
Scoring rules:
3 = Excellent The sentiment polarity in the sentiment explanation is completely consistent with the tuple;
Sentiment expression is clear, complete, and unambiguous.
2 = Good The sentiment explanation contains sentiment polarity consistent with the tuple;
But the expression is not clear enough, or there is slight ambiguity;
Overall sentiment still focuses on the tuple's sentiment.
1 = Poor The sentiment explanation involves the tuple event, but the sentiment polarity is clearly inconsistent with the tuple;
Or the sentiment expression is ambiguous, making it difficult to determine the attitude; Resulting in low sentiment consistency with the tuple.
0 = Irrelevant/Error The sentiment explanation does not express sentiment for the tuple event at all;
Or conveys sentiment polarity opposite to the tuple, causing serious conflict.Output Format:
The output must use JSON format, with a fixed format of: {"score": X, "explanation": "Your explanation"}
Core Principles
Main focus: Whether the sentiment polarity (positive/negative/neutral) of the target event in the sentiment explanation is consistent with the sentiment polarity in the tuple.If the sentiment polarity in the sentiment explanation is completely consistent with the tuple → High score;If the sentiment polarity expressed in the sentiment explanation is inconsistent with, missing, or ambiguous compared to the tuple → Reduce score;If the sentiment explanation does not involve relevant sentiment at all → Record as the lowest score.
Example:
Example 1:
Quadruple: ["2015", "Newmi Technology", "company listing", "positive"] Tuple Sentiment: positive
Sentiment Explanation: Newmi Technology was listed on the NEEQ in 2015, successfully raised funds, and received enthusiastic market response.
Output: {{"score": 3, "explanation": "The sentiment explanation conveys clear positive sentiment, consistent with the tuple's positive sentiment, with complete expression."}}
Example 2:
Quadruple: ["2015", "Newmi Technology", "company listing", "positive"] Tuple Sentiment: positive
Sentiment Explanation: Newmi Technology was listed on the NEEQ in 2015, but the market response was mediocre.
Output: {{"score": 2, "explanation": "The sentiment explanation generally tends to be positive, but the tone is not positive enough, with slight ambiguity, thus scoring as good."}}
Example 3:
Quadruple: ["2015", "Newmi Technology", "company listing", "positive"] Tuple Sentiment: positive
Sentiment Explanation: Newmi Technology was listed on the NEEQ in 2015, but suffered a loss of 220 million yuan in 2019, with declining market share.
Output: {{"score": 1, "explanation": "The sentiment explanation includes the listing event, but overall conveys negative sentiment, inconsistent with the tuple's positive sentiment."}}
Example 4:
Quadruple: ["2015", "Newmi Technology", "company listing", "positive"] Tuple Sentiment: positive
Sentiment Explanation: Newmi Technology suffered a loss of 220 million yuan in 2019, with declining market share.
Output: {{"score": 0, "explanation": "The sentiment explanation does not express sentiment for the listing event; the sentiment in the explanation is irrelevant to the tuple sentiment."}}
Please score the new input according to the above rules and examples.

Input

Quadruple: {event_tuple}
Quadruple Sentiment: {event_tuple[3]}
Sentiment Explanation: {explanation}

Figure 13: Prompts for event sentiment consistency evaluation. We show examples for the model to better comprehend the evaluation.

Event Subject/Object Consistency Evaluation Prompt

System Prompt

Task
You are a rigorous reviewer responsible for evaluating the "financial event subject/object consistency" between 【quadruple】 and 【Event sentiment explanation】. Please score the input according to the following rules and provide an explanation.
Scoring rules:
3 = Excellent: The financial entities in the explanation are completely consistent with the tuple entities, and all sentiment information is directly related to this entity; reasonable details related to this entity (such as time, amount, project name, business background, etc.) can be included, which will not affect the score; sentiment information about other financial entities unrelated to the tuple entity must not be introduced.
2 = Good: Contains and clearly expresses the sentiment of the target entity; but simultaneously introduces secondary sentiment about unrelated entities, with slightly distracted attention.
1 = Poor: The target entity appears but its sentiment expression is vague/unclear; accompanied by sentiment about unrelated entities, causing obvious interference or focus shift. Sentiment information about other financial entities unrelated to the tuple entity also appears, causing interference.
0 = Irrelevant/Error: The explanation does not mention the tuple entity; or the financial entity involved is completely inconsistent with the tuple entity.
Output Format:
The output must use JSON format, with a fixed format of:
{"score": X, "explanation": "Your explanation"}
Example:
Example 1:
Quadruple: ["Saudi Aramco", "2019", "company listing", "positive"]
Explanation: Saudi Aramco listed in 2019, becoming the largest IPO at that time.
Output: {"score": 3, "explanation": "The explanation completely corresponds to the tuple entity, with clear and complete sentiment expression, without introducing additional financial entities."}
Example 2:
Quadruple:["Saudi Aramco", "2019", "company listing", "positive"]
Explanation: Saudi Aramco listed in 2019, becoming the largest IPO at that time. However, Ant Group plans to raise over $34 billion through listings in Shanghai and Hong Kong, which will exceed Saudi Aramco.
Output: {"score": 2, "explanation": "Contains the tuple entity and fully expresses its sentiment, but simultaneously introduces sentiment information about the additional financial entity Ant Group."}
Example 3:
Quadruple: ["Saudi Aramco", "2019", "company listing", "positive"]
Explanation:Ant Group plans to raise over $34 billion through listings in Shanghai and Hong Kong, which will exceed Saudi Aramco.
Output: {"score": 1, "explanation": "Although Saudi Aramco is mentioned, its listing sentiment information is not fully expressed, and sentiment information about the additional financial entity Ant Group is also introduced."}
Example 4:
Quadruple:["Saudi Aramco", "2019", "company listing", "positive"]
Explanation:Tesla plans to expand its Gigafactory in Shanghai.
Output: {"score": 0, "explanation": "The explanation does not mention the tuple entity Saudi Aramco, and the financial entities are completely inconsistent."}

Input

Event Sentiment Quadruple :{tuple}

Event Sentiment Explanation: {explanation}

Figure 14: Initial prompts for event subjec/object consistency evaluation. We show examples for the model to better comprehend the evaluation.

Overall Consistency Evaluation Prompt

System Prompt

Task
You are a rigorous reviewer responsible for comprehensively evaluating the overall consistency of the 【Sentiment Explanation】 based on three fine-grained consistency scores, and assessing its logical coherence.
Evaluation Dimensions
Dimension 1: Fine-Grained Consistency Comprehensive Performance (Weight: 50%)
Dimension 2: Logical and Linguistic Coherence Assessment (Weight: 50%)
Comprehensively evaluate the logical structure and language expression of the sentiment explanation:

###Evaluation Elements:
(1) Information Completeness: Whether it fully covers the key elements of the tuple event
(2) Argument Rationality: Whether supporting information has reasonable relevance to the core event
(3) Expression Fluency: Whether the language is natural and fluent, without abrupt transitions
(4) Transition Naturalness: Whether transitions between sentences are natural
(5) Sentiment Consistency: Whether the overall sentiment tone maintains consistency with the tuple sentiment
###Scoring Criteria (1-5 points):
5 points (Excellent): Logical structure is rigorous and clear, language is extremely fluent, arguments are sufficient, no flaws whatsoever
4 points (Good): Logic is basically clear, language is fluent, occasional minor flaws that don't affect understanding
3 points (Average): Logic is basically coherent, language is basically fluent, with slight jumps or lack of fluency
2 points (Poor): Logic is chaotic, language is not fluent, with obvious contradictions or breaks
1 point (Terrible): Logic is severely chaotic, language is severely unfluent, incomprehensible
Comprehensive Score = (Fine-Grained Consistency Average Score ÷ 3 × 0.5 + Logical Coherence Score ÷ 5 × 0.5) × 5
##Output Format:
{
"overall_score": X.X,
"comprehensive_evidence": "Comprehensive evidence stating the reasons for the score, contribution of each dimension, and improvement suggestions",
}

Input

Event Quadruple: {tuple}
Sentiment Explanation: {explanation}
Three Fine-Grained Consistency Score Results
1. Sentiment Consistency Score
Score: {sentiment score} / 3
Reason: {evidence}
2. Event Consistency Score
Score: {event score} / 3
Reason: {evidence}
3. Entity Consistency Score
Score: {entity score} / 3
Reason: {evidence}

Figure 15: Prompts for overall consistency evaluation.

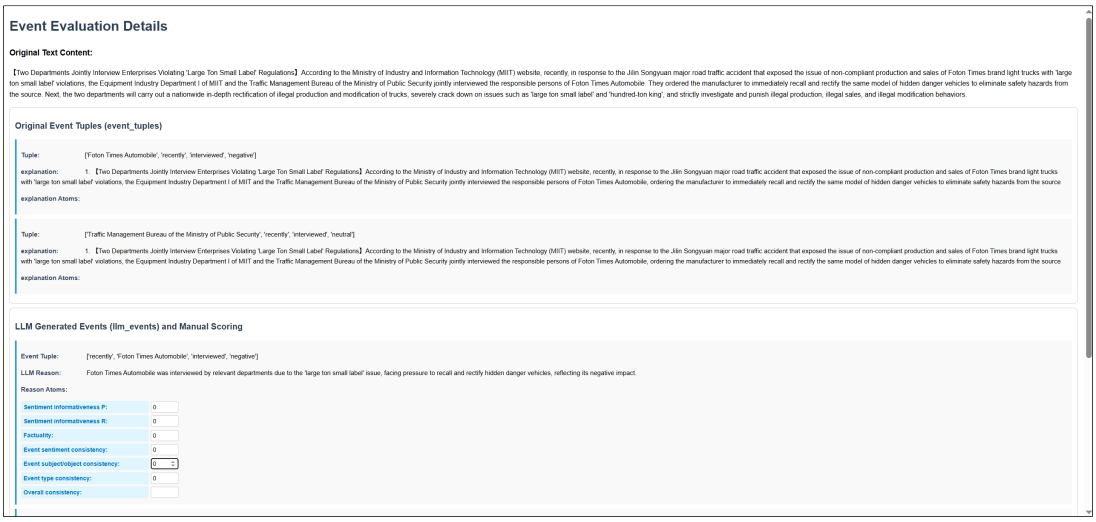


Figure 16: The manual evaluation system and examples.